Quiz: BigTable

Test your knowledge of BigTable by completing this quiz.

Quiz ?

Statement

In terms of CAP theorem, BigTable is --- system.

Select Multiple Options

False ⊗
☐ AP

Correct ✓
☑ CP

False ⊗
☐ CA

Check Answer

Correct Answer ✓

Quiz ?

Statement

BigTable is indexed by a unique key which is made up of

Select Multiple Options

False ⊗
☐ Row Key

False ⊗
☐ Primary Key

Correct ✓
☑ Row key, Column Key, and Timestamp

False ⊗
☐ Master Key and Column Key

Check Answer

Correct Answer ✓

Quiz ?

Statement

How does Bigtable split a table to store on to multiple servers?

Select Multiple Options

False ⊗
☐ By column family, with all columns from that part of a column family placed on the same servers.

Correct ✓
☑ By rows, with a contiguous set of rows placed on different servers.

False ⊗
☐ By columns, with a contiguous set of columns placed on different servers.

False ⊗
☐ By locality group, with all columns from that part of the locality group placed on the same servers.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Why is choosing a good row key important in BigTable?

Select Multiple Options

False ⊗
☐ To avoid hotspots by distributing contiguous rows.

Correct ✓
☑ To take advantage of the locality of adjacent rows.

False ⊗
☐ To be able to identify a row uniquely.

False ⊗
☐ To search for an arbitrary row based on its key.

Check Answer

Correct Answer ✓

Quiz ?

Statement

What is the unit of distribution and load-balancing in BigTable?

Select Multiple Options

False ⊗
☐ SSTable

Correct ✓
☑ Tablet

False ⊗
☐ MemTable

False ⊗
☐ Column family

Check Answer

Correct Answer ✓

Quiz ?

Statement

What are the different compaction strategies that are used in BigTable? Select all that apply.

Select Multiple Options

Correct ✓
☑ Converting MemTables to SSTables

False ⊗
☐ Merging all SSTables based on their time of creation

Correct ✓
☑ Merging some SSTables and MemTables into a new SSTable

Correct ✓
☑ Merging all SSTables into a single SSTable

Check Answer

Correct Answer ✓

Quiz ?

Statement

What are the responsibilities of the master server? Select all that apply.

Select Multiple Options

Correct ✓
☑ Handling metadata operations

Correct ✓
☑ Garbage collection of underlying files in GFS

Correct ✓
☑ Assigning Tablets to Tablet servers

False ⊗
☐ Notifying clients about the existence of new Tablet servers

Check Answer

Correct Answer ✓

Quiz ?

Statement

BigTable uses which of the following schemes to enhance read performance? Select all that apply.

Select Multiple Options

Correct ✓
☑ Bloom filters

False ⊗
☐ Column family

Correct ✓
☑ Locality Groups

Correct ✓
☑ Cache

Check Answer

Correct Answer ✓

Quiz ?

Statement

How can we describe BigTable's MemTable?

Select Multiple Options

False ⊗
☐ It stores data mutations in the Master's memory that are yet to be sent to a tablet server.

False ⊗
☐ It caches the frequently accessed rows stored on a Tablet server.

False ⊗
☐ It caches the frequently accessed rows stored on the Master.

Correct ✓
☑ It stores recently committed changes to BigTable stored on a Tablet server.

Check Answer

Correct Answer ✓

Quiz ?

Statement

What is the use of Metadata Tablets?

Select Multiple Options

Correct ✓
☑ To enable clients to find the tablet server responsible for a row.

False ⊗
☐ To cache the Tablet to Tablet server mapping on the Master.

False ⊗
☐ To store access control permissions of Tablets.

False ⊗
☐ To help the Master for balancing the load distribution of tablets among servers.

Check Answer

Correct Answer ✓

← Previous
Summary: BigTable

Next →
Mock Interview: BigTable

☐ Mark as Completed